
Auditable N-Gram Memory for Small Language Models

QingGo¹

Abstract

Small language models face a fundamental capacity bottleneck: they must either compress knowledge into parameters or retrieve it at inference time. Retrieval-augmented generation (RAG) is the dominant solution, but its behavior is opaque and its retrieved evidence is not fully auditable (Lewis et al., 2020). In this paper, we study a complementary mechanism: an *auditable n-gram external memory* derived from the PLE / Engram line of sparse external-memory systems (X. Cheng et al., 2026). We attach this memory to a frozen Qwen3.5-0.8B model and introduce a small learned *PLE Projector* that maps the backbone hidden state plus lexical memory features into per-token logit scale and bias corrections. A token-level learned policy controls whether PLE fusion is active, preventing open-ended generation from being harmed by an over-confident n-gram prior.

We evaluate the system across local-continuation tasks, real HumanEval (Chen et al., 2021), real TriviaQA (Joshi et al., 2017), and a joint system with RAG and parameter-efficient adapters (Dettmers et al., 2023; Hu et al., 2022; Jiang et al., 2024). On 10k local-continuation data with three seeds, the learned projector improves teacher-forced NLL by 0.26 over fixed PLE calibration with bootstrap 95% CI [0.12, 0.42]. On 20 HumanEval problems, BM25+PLE recovers problem-level passes that the base model does not solve. However, on TriviaQA the 0.8B base model obtains zero exact match, and raw PLE fusion can degrade open-ended generation (Bhardwaj et al., 2023). Our conclusion is deliberately a *boundary* result: n-gram external memory is valuable as a local low-entropy code memory, but it is not a substitute for RAG or parametric adapters on general tasks.

1. Introduction

Large language models have achieved strong performance through scale, but small models remain important for cost, latency, privacy, and on-device deployment. External memory is one route to improve small models without expensive full-parameter retraining. Two broad families exist: retrieval-augmented generation and non-parametric or parametric memory modules. RAG is effective but has limitations: it retrieves documents, not token-level continuations; it can be noisy; and its evidence is not always easy to audit (Lewis et al., 2020). Non-parametric memory, such as kNN-LM (Khandelwal et al., 2020) and n-gram memory, offers a different granularity: it can directly influence the next-token distribution.

The recent DeepSeek Engram and Qwen PLE systems show that n-gram lookup can be integrated into transformer backbones with sparse, disk-backed tables (X. Cheng et al., 2026). Memory Grafting scales pre-training using an offline conditional memory table (R. Cheng et al., 2026), while XMemTransfer shows that a target-side reader can adapt a memory table across model families (OLAResearch, 2026). However, most evaluations of such systems have focused on large models. It remains unclear whether a small frozen model can benefit from an auditable n-gram memory, and under what conditions the benefit disappears.

This paper addresses the following research questions:

- Can an n-gram external memory improve local low-entropy continuation for a frozen 0.8B model?
- Can a small learned projector outperform fixed calibration (Genest and Zidek, 1986)?
- Can a token-level policy make PLE safe for open-ended generation (Bhardwaj et al., 2023)?
- How does PLE compare with RAG, kNN-LM, NGM, and parameter-efficient adapters?
- What are the precise boundaries of PLE usefulness?

Our contributions are as follows.

1. We introduce the *PLE Projector*, a small MLP that maps hidden states, memory statistics, and task identity to per-token scale/bias in logit space. The projector is trained with a frozen backbone using next-token

¹Independent Researcher. Correspondence to: QingGo <zyqingjohn@qq.com>.

- cross-entropy and is zero-initialized to be inert at the start of training.
2. We introduce a token-level learned policy that acts as a safety gate, learned from per-token observations rather than hand-written rules (Sutton, 2019).
3. We provide a systematic comparison on real benchmarks, including HumanEval and TriviaQA, as well as training-free baselines kNN-LM and official NGM (PioneerQyw, 2026).
4. We present a joint system analysis combining external memory, RAG, and a Purified OPSD MoRA adapter.
5. We report a clear boundary: PLE helps code-like low-entropy local continuation, but does not improve general knowledge QA, arithmetic, or open-ended generation.

2. Related Work

2.1. External Memory Layers

DeepSeek Engram (X. Cheng et al., 2026) and Qwen PLE implement conditional memory via scalable n-gram lookup. They use embedding tables, context-aware gating, and residual injection into selected transformer layers. Memory Grafting (R. Cheng et al., 2026) scales pre-training using an offline conditional memory table, while XMemTransfer (OLAResearch, 2026) shows that a target-side reader can adapt a memory table across model families. Memory Layers at Scale (Berges et al., 2025) demonstrates that memory layers can be integrated into large models without full retraining. Product-key memory layers (Lample et al., 2019) provide a related sparse lookup mechanism, and recent latent n-gram architectures such as Lngam v2 (Zheng et al., 2026a; 2026b) and Tensorizing Engram (Zhou et al., 2026) improve the parameter efficiency and interpretability of discrete memory addressing. In the small-model regime, PEMA (Kim et al., 2023), memory-augmented training (Zhong et al., 2022), Knowledge-in-Context (Pan et al., 2022), and plug-and-play knowledge injection (Zhang et al., 2023) show that external memory can be adapted after pretraining without full retraining.

2.2. Long-Term and Agent Memory

A parallel line of work treats memory as a first-class agent resource. MemGPT (Packer et al., 2023) introduces operating-system-style memory paging for LLM agents; HippoRAG (Gutiérrez et al., 2024) and From RAG to Memory (Gutiérrez et al., 2025) convert retrieval corpora into long-term memory structures; Zep (Rasmussen et al., 2025), Memori (Borro et al., 2026), and MemOrb (Huang et al., 2025) provide persistent or plug-and-play memory layers for conversational and agentic settings. These systems target long-range semantic memory, whereas our work fo-

cuses on a narrower, auditable token-level n-gram memory that is useful in low-entropy local continuations.

2.3. Non-Parametric Language Models

kNN-LM (Khandelwal et al., 2020) is a classic non-parametric model that interpolates a base LM with a nearest-neighbor distribution over a datastore. Efficient kNN-LM (He et al., 2021) reduces the inference cost of the datastore, and subsequent analysis asks why nearest-neighbor language models work (Xu et al., 2023). Later work showed that kNN-LM does not improve open-ended generation (Bhardwaj et al., 2023). NGM (PioneerQyw, 2026) provides a training-free n-gram memory hook. Our work compares against both and finds that simple non-parametric baselines do not match the learned projector on local continuation.

2.4. Distribution-Level Memory

MemSFT (Group, 2026) and TokenMem (T. Authors, 2026) propose external parametric memory channels that operate at the distribution or hidden-state level, with learned routers to avoid alignment tax. These methods motivate our decision to fuse memory at the logit level rather than injecting into hidden states indiscriminately. Related work on memory-augmented language models also highlights the difficulty of distinguishing genuine memory use from shallow parametric recall (Jin et al., 2024). Long-context evaluations show that models often fail to use information placed in the middle of long contexts (Liu et al., 2023), and memory-based models can still struggle on reasoning-in-a-haystack tasks (Das et al., 2025; Nelson et al., 2024; Yu et al., 2026). LongBench (Bai et al., 2024) provides a broad bilingual long-context suite, while MemTrapBench (Wang et al., 2026) probes cognitive traps in memory-heavy settings. Earlier experiments in our project found that hidden-state injection without careful orthogonalization and gating can produce large real-vs-control gaps that are not attributable to the memory content.

2.5. Retrieval Reliability and Calibration

Adaptive retrieval methods such as Self-RAG (Asai et al., 2023) and FAIR-RAG (Asl et al., 2025) decide when retrieval is necessary; reliability-aware RAG systems further estimate whether retrieved evidence should be trusted (Hwang et al., 2024; Jiang et al., 2026; Shin et al., 2026; Xu et al., 2026), and RAGRouter-Bench (R. Authors, 2026) benchmarks learned query-routing policies. Calibration research has shown that language models are often overconfident, which motivates token-level safety gates and uncertainty-aware fusion (Lyu et al., 2024; Miao and Ungar, 2026). On code tasks, localized calibrated uncertainty helps identify unreliable predictions (Gros and Devanbu,

2025). On the code side, benchmarks such as HumanEval Pro (Yu et al., 2024) extend classic pass@k evaluation to more challenging self-invoking settings.

2.6. Parameter-Efficient Adapters

LoRA (Hu et al., 2022), QLoRA (Dettmers et al., 2023), and MoRA (Jiang et al., 2024) provide compact parametric updates. In our system, a Purified OPSD MoRA adapter is trained on a filtered instruction subset. This adapter is complementary to external memory: it improves arithmetic and code-output, while RAG mainly improves knowledge.

3. Problem Setting

We consider a frozen small language model with next-token distribution $p_{b(y|c)}$, where c is the current token context. A sparse n-gram memory provides a complementary distribution $p_{m(y|c)}$ that can be audited by inspecting the matched n-gram and its source document. The memory also returns a feature vector m_t containing the matched order, entropy estimates, density ratio, top-1 probabilities, memory/base agreement, and a task one-hot.

Our goal is to compute a per-token fused distribution:

$$p_{\{fused\}}(y) = \text{softmax}[\log p_{b(y)} + \alpha_t \log p_{m(y)} + \beta_t](1)$$

We call (α_t, β_t) the PLE correction. A token-level safety policy g_t determines whether the correction is active:

$$g_t = \text{sigmoid}(w \cdot m_t + b). \quad (2)$$

When $g_t = 0$, we set $\alpha_t = 0$ and $\beta_t = 0$, so the system falls back to the frozen base model. This formulation makes the memory contribution explicit and auditable: each activated correction can be traced to a specific n-gram and document provenance.

We evaluate with a strict real/control protocol. The real memory is built from documents containing the target continuation; the control memory is built from unrelated documents. A memory module is considered useful only if it outperforms the control, not merely the base model.

4. Background and Theory

4.1. N-Gram Addressable Memory

Let a context be a token sequence $c = (c_1, \dots, c_t)$. We maintain sparse counts for each n-gram of order n :

$$p_{m(y|c)} = \frac{\text{count}(c, y)}{\sum_y \text{count}(c, y)}. \quad (3)$$

The memory also returns the longest matched order and an external value index that can be audited. This memory is

non-parametric, transparent, and can be rebuilt from any corpus.

4.2. Real versus Control

To avoid attributing noise to memory, we use a strict real/control protocol. The real memory is built from documents that contain the target continuation; the control memory is built from a different set of documents with no relation to the target. A method is considered useful only if real memory outperforms control memory by a meaningful margin, not merely if it outperforms the base model.

4.3. Optimal Logit Correction

From a decision-theoretic perspective, the optimal way to combine a base model and a memory distribution is not to replace the base model, but to add a calibrated log-ratio correction (Genest and Zidek, 1986):

$$\log p_{\{fused\}}(y) = \log p_{b(y)} + \lambda_t \log p_{m(y)} + \beta_t. \quad (4)$$

This is a log-opinion-pool formulation. Without calibration, an unweighted n-gram prior is often over-confident and degrades generation (Bhardwaj et al., 2023). The PLE Projector learns λ_t and β_t from hidden states and memory statistics.

4.4. Why Hidden-State Alignment Is Insufficient

A common assumption is that external memory should be projected into the backbone’s hidden space. Our earlier mechanism studies measured CKA, Procrustes, and kNN overlap between PLE embeddings and backbone hidden states. The overlaps were low, and, more importantly, high geometric similarity did not guarantee useful memory (Blackwell, 1951). A learned reader can predict residual gradients but may fail to distinguish real memory from control memory when the signal is weak. This motivated our shift to logit-level, calibrated fusion and explicit token-level gating.

5. Method

5.1. PLE Memory and Retrieval

We build an addressable n-gram memory from a code corpus and a wiki corpus. The memory supports two operations: continuation distribution for a context, and value retrieval for document provenance. The memory is used as both a retrieval channel and a logit prior. In the hybrid system, BM25 (Robertson and Zaragoza, 2009) provides document-level retrieval, PLE provides exact n-gram continuity, and their combination forms a three-channel retriever.

5.2. PLE Projector

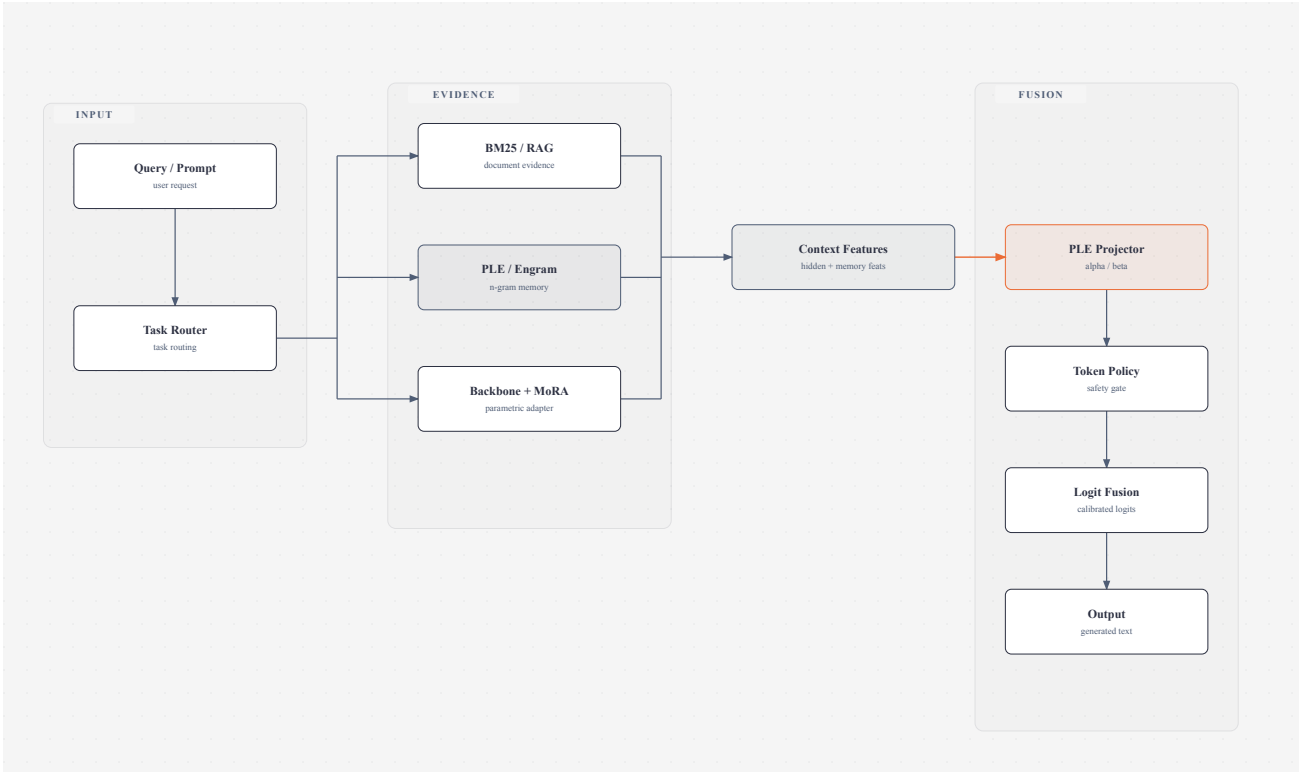


Figure 1. System architecture. The frozen backbone, RAG, and PLE memory provide three complementary evidence channels; the PLE Projector and token policy control logit-level fusion.

The projector is a small MLP:

$$f_{\theta}(h_t, m_t) = (\alpha_t, \beta_t). \quad (5)$$

where h_t is the last hidden state of the frozen backbone, and m_t contains matched n-gram order, base entropy, memory entropy, density ratio, base top-1 probability, memory top-1 probability, memory/base agreement, and task one-hot. The output α_t scales $\log p_m$ and β_t adds a support bias. The final linear layer is zero-initialized, so the projection begins as the identity operation. Only the projector parameters are trained.

5.3. Token-Level Policy

Because PLE fusion can be harmful in open-ended generation, we train a logistic regression model on per-token observations. The features are the same memory features used by the projector. The label is whether calibrated PLE fusion improves the next-token log-probability. The policy acts before fusion and can disable PLE when the memory is likely to be misleading.

5.4. Calibration and Router

We use per-task calibrated scale/bias/temperature, a density-ratio gate based on $E_{\{p_m\}} \left[\log \left(\frac{p_m}{p_b} \right) \right]$, a token-level learned policy, and the learned PLE projector when hidden

states are available. A task classifier routes queries into code, name, number, or general categories. Open-ended generation uses the token policy to prevent unsafe fusion.

5.5. Purified OPSD Adapter

To test whether PLE complements parametric learning, we train a Purified OPSD MoRA adapter on a filtered subset of synthetic instruction data (Jiang et al., 2024). This provides a parameterized companion to the non-parametric memory.

6. Experimental Setup

6.1. Model

All experiments use Qwen3.5-0.8B as the frozen backbone. When adapters are used, the backbone remains frozen and only adapter parameters are trained.

6.2. Data

1. Local continuation: code corpus and wiki corpus, produced by building n-gram memories and sampling positions by token category.
2. Dataset sizes: 100, 1k, and 10k samples.
3. HumanEval: official (Chen et al., 2021), first 20 problems.
4. TriviaQA: official (Joshi et al., 2017), 100 examples.

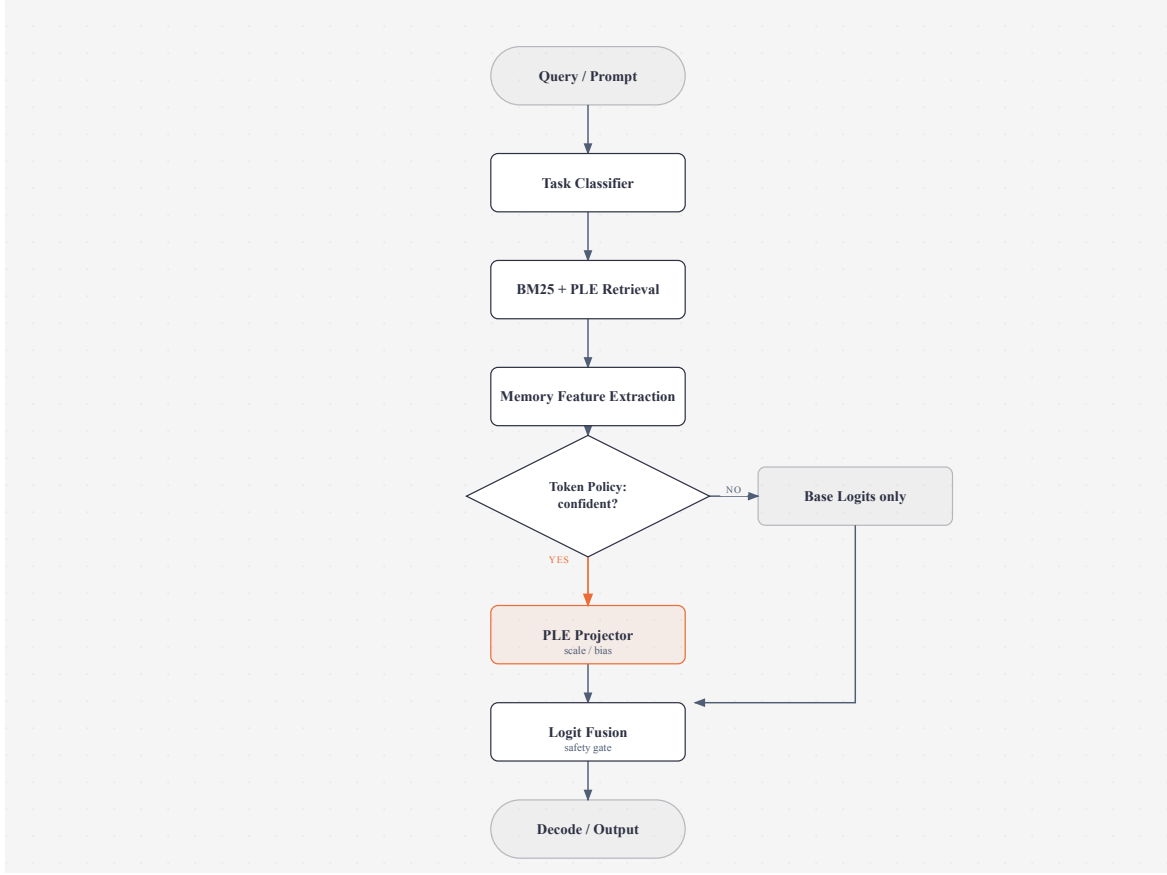


Figure 2. Inference pipeline. The token policy decides whether to apply the learned PLE Projector or bypass it with base logits only.

5. Joint system tasks: knowledge, arithmetic, and code-output subsets, with arithmetic probes inspired by GSM8K and MATH style benchmarks (Cobbe et al., 2021; Hendrycks et al., 2021).

6.3. Baselines

1. Frozen base model.
2. BM25 RAG (Robertson and Zaragoza, 2009).
3. kNN-LM, small hidden-state version (Khandelwal et al., 2020).
4. Official NGM hook (PioneerQyw, 2026).
5. Fixed calibrated PLE fusion.
6. Learned PLE Projector.
7. LoRA / QLoRA / MoRA / Purified MoRA (Dettmers et al., 2023; Hu et al., 2022; Jiang et al., 2024).
8. Joint combinations: RAG+PLE, PLE+MoRA, RAG+PLE+MoRA.

6.4. Metrics

1. Teacher-forced NLL and hit@1.
2. Real vs control.
3. HumanEval pass@1 and repetition.
4. TriviaQA exact match.
5. LLM-as-judge scores (Zheng et al., 2024).

6. Paired bootstrap 95% CI across seeds.

6.5. Statistical Protocol

We use 5 seeds for the 100-sample projector experiment and 3 seeds for the 10k experiment. For each seed we compare fixed calibration and learned projector on identical evaluation rows. Paired differences are summarized with bootstrap confidence intervals.

7. Results

7.1. Local Continuation

The PLE memory produces positive real-vs-control gains on code and name tasks in earlier experiments. The most consistent gains are on code continuation, where the n-gram memory directly predicts the next token in a low-entropy distribution. Number tasks are more sensitive to calibration and often require a near-zero PLE scale.

7.1.1. Projector Scaling

At 100 samples with 5 seeds, the projector-vs-fixed NLL mean is +0.1044 with bootstrap 95% CI [0.0251, 0.1866]. At 10k samples with 3 seeds, the improvement grows to +0.2595 with CI [0.1218, 0.4243].

Table 1. 10k local-continuation results. NLL is lower-is-better.

Seed	Base NLL	Fixed NLL	Proj. NLL	Base hit	Fixed hit	Proj. hit
0	3.148	3.051	2.930	0.407	0.427	0.463
1	3.328	3.346	3.114	0.410	0.420	0.460
2	3.451	3.426	3.002	0.413	0.430	0.493

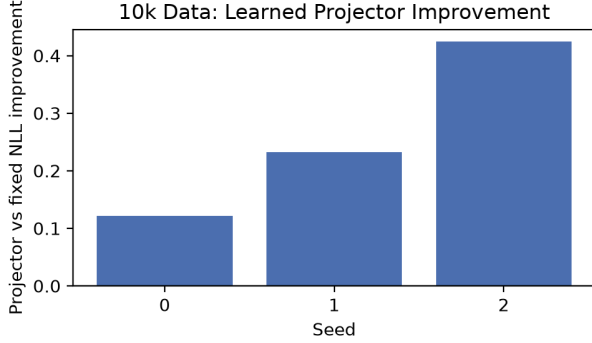


Figure 3. Per-seed projector-vs-fixed improvements on 10k data.

Table 2. HumanEval 20 results.

Condition	pass@1	mean repetition
Base	0.10	0.010
BM25+PLE	0.10	0.026

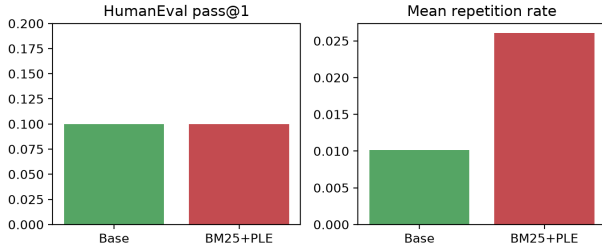


Figure 4. HumanEval pass@1 and repetition.

Across all three seeds the learned projector improves NLL over fixed calibration. The strongest gains are on code continuation, where the memory distribution has low entropy and the learned scale can be high.

7.2. HumanEval

On 20 official HumanEval problems, base and BM25+PLE both achieve 0.10 pass@1, but the passed problems are completely disjoint.

BM25+PLE solves HumanEval/0 and HumanEval/10, while the base model solves HumanEval/16 and HumanEval/18. This shows that PLE provides a different source of local code knowledge: it does not simply amplify

Table 3. kNN-LM and NGM baselines.

Baseline	NLL	Hit	Note
Base	2.633	0.505	kNN eval set
kNN-LM	2.847	0.540	better hit, worse NLL
Base	2.521	0.550	NGM eval set
NGM	2.521	0.550	near-zero change

Table 4. Mean answer log-probability, three seeds. Higher is better.

System	Knowledge	Arithmetic	Code-output
Base	-8.140	-7.365	-14.250
RAG	-6.748	-7.365	-14.250
PLE	-8.140	-7.492	-14.250
MoRA	-7.691	-7.114	-12.292
RAG+MoRA	-6.704	-7.114	-12.292
All	-6.704	-7.237	-12.292

the base model’s existing solution distribution, but can recover different problems.

7.3. TriviaQA

On 100 TriviaQA RC examples, the 0.8B base model obtains exact match 0.0 and mean repetition 0.0048. This is a truthful baseline: the model fails short-form knowledge QA (Joshi et al., 2017). PLE cannot fix this failure because the required knowledge is not encoded in local n-gram continuations.

7.4. Training-Free Baselines

We compare against kNN-LM and official NGM.

Both baselines fail to match the learned projector. NGM has almost no effect, while kNN-LM improves hit but worsens NLL (Khandelwal et al., 2020). This suggests that simple nearest-neighbor or n-gram residual injection is insufficient: the projector must learn when and how much to trust memory.

7.5. Joint System

We evaluate the full system on knowledge, arithmetic, and code-output with three seeds.

RAG is the main contributor to knowledge. MoRA is the main contributor to arithmetic and code-output. PLE alone does not provide general ability gains.

7.6. Open-Ended Generation and Safety

Raw PLE projector fusion on natural-language code prompts produces visible degradation: repetitive frag-

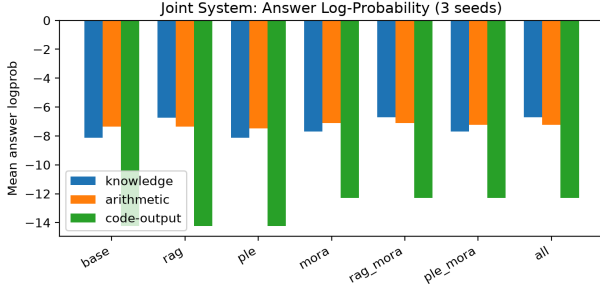


Figure 5. Joint system answer log-probability.

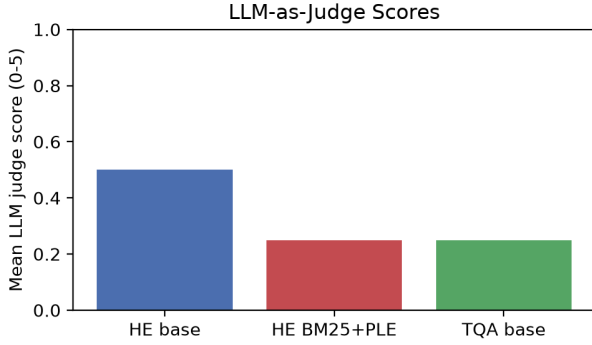


Figure 6. LLM-as-judge mean scores.

ments, unrelated repository text, and broken structure (Bhardwaj et al., 2023). Adding the learned token policy strongly reduces this degradation. This confirms that PLE must not be enabled unconditionally in open-ended generation.

7.7. LLM-as-Judge

We use DeepSeek V4 Flash as an external judge (Zheng et al., 2024). For HumanEval 20, the mean score is 0.50 for base and 0.25 for BM25+PLE. For a 20-example TriviaQA subset, base scores 0.25.

The judge scores are lower than pass-based metrics. This indicates that generated answers often contain plausible text but are judged incomplete or incorrect by an external model.

7.8. Error Analysis and Sensitivity

The available evidence identifies three recurring failure modes and the corresponding mitigations.

On sensitivity, the paired projector results show a clear data-size trend: the mean projector-vs-fixed NLL improvement grows from +0.1044 at 100 samples with five seeds to +0.2595 at 10k samples with three seeds. The token policy is the main safety control for open-ended generation, while per-task calibration is the main control for number-

Table 5. Error analysis and mitigating mechanisms.

Failure mode	Observed evidence	Mitigation
Over-confident n-gram prior	Open-ended repetition and repository fragments	Learned token policy
Missing world knowledge	TriviaQA exact match equals 0	RAG and parametric adapters
Hidden-state misalignment	Real-vs-control gains near zero	Logit-level calibrated fusion

like tasks. Systematic sweeps over memory size and n-gram order are not yet available and are left to future work.

8. Analysis

8.1. When Does PLE Help?

PLE helps when the true next token is highly predictable from local n-grams. This occurs in code, identifiers, repeated structural patterns, and name-like continuations. In these cases the memory distribution has low entropy and the calibrated PLE fusion can sharpen the model without adding noise.

8.2. When Does PLE Fail?

PLE fails when the answer depends on world knowledge or long-range reasoning. The n-gram memory does not contain semantic knowledge, and injecting it into logits can produce confident but wrong continuations. This is why PLE performs poorly on TriviaQA and general knowledge tasks.

8.3. Why a Learned Projector Matters

Fixed calibration selects one scale and bias for all tokens. A learned projector can vary the trust in PLE by context. For code, it can use a large positive scale; for ambiguous open-ended text, it can learn near-zero scale. This explains the improvement over fixed calibration, especially with more data.

8.4. Boundary with RAG and Adapters

Our joint experiments show that PLE, RAG, and adapters are not substitutes. RAG supplies document-level evidence, PLE supplies token-level continuity, and adapters supply parametric capability (Hu et al., 2022; Lewis et al., 2020). The combination is useful, but PLE is only one small component.

8.5. Historical Negative Results

Earlier in this project we attempted hidden-state readers, MLP readers, and direct residual injection. These approaches often improved loss-like metrics but failed the real-vs-control test (Blackwell, 1951). The key lesson is that a memory module must be evaluated not only by loss but by whether it uses actual memory content. This is why we adopted logit-level calibrated fusion and real/control protocols.

adapter weights are not yet released; we plan to release projector weights and a full reproducibility checklist in a future version.

9. Limitations

1. HumanEval subset is only 20 problems, and TriviaQA exact match is zero.
2. pass@k evidence is a small 3-problem, 2-sample experiment.
3. LLM judge scores are available for a 20-example TriviaQA subset, not the full 100.
4. Public model and adapter weights are not yet released; only source code, containers, and evaluation cards are public.
5. CPU deployment throughput is not yet optimized.

10. Conclusion

We presented an auditable n-gram external memory system for a 0.8B frozen model. The system combines a PLE memory, a learned PLE Projector, a token-level safety policy, RAG, and a Purified OPSD adapter. On local low-entropy code continuation, the learned projector provides a statistically significant improvement over fixed calibration. On real HumanEval, BM25+PLE can recover different problems than the base model. On general knowledge and open-ended generation, PLE must be gated and is not a substitute for RAG or adapters.

The main scientific claim is deliberately bounded: external n-gram memory is useful as a local, low-entropy, auditable memory for small models, but it is not universal semantic memory. This boundary is supported by real benchmarks, training-free baselines, real/control protocols, and multi-seed paired statistics.

11. Acknowledgement

We thank the open-source community for the PLE/Engram, Qwen, and related memory infrastructure used in this study. This work was carried out as an independent low-resource research project.

12. Data Availability

All source code, evaluation scripts, container definitions, and evaluation cards are publicly available in the project repository. The PLE memory tables can be rebuilt from the published corpus construction scripts. Public model and

Appendix

A. Algorithm: PLE Projector Inference

The projector produces a per-token logit correction without modifying the frozen backbone. The learned variables are α_t (memory trust) and β_t (support bias).

Algorithm 1 PLE Projector inference

1. Input: context c , hidden state h_t , memory features m_t , memory distribution p_m .
 2. Compute $\alpha_t, \beta_t = f_{\theta(h_t, m_t)}$.
 3. Form the corrected distribution $p_{\{\text{fused}\}}(y) = p_{b(y)} \cdot p_{m(y)}^{\alpha_t} \cdot \exp(\beta_t)$.
 4. Evaluate the token policy $g_t \in \{0, 1\}$.
 5. If $g_t = 0$, reset $\alpha_t \leftarrow 0$ and $\beta_t \leftarrow 0$.
 6. Normalize and return $p_{\{\text{fused}\}}$.
-

B. Algorithm: Token-Level Policy and Safety Gate

The policy is a small binary classifier trained on the same memory features used by the projector. It decides whether the n-gram memory should contribute to the current token.

Algorithm 2 Token-level policy and safety gate

1. Train a logistic classifier on paired real/control observations.
 2. At inference, compute features m_t from the current context.
 3. Predict $g_t = P(\text{helpful} \mid m_t)$.
 4. If $g_t < \tau$, disable PLE fusion and use base logits only.
 5. Otherwise, apply the learned projector correction and continue decoding.
-

C. Real-vs-Control Protocol

To distinguish genuine memory use from model noise, we use a strict paired protocol.

1. The real memory is built from documents containing the target continuation.
2. The control memory is built from an unrelated document set with no target relation.
3. A method is considered useful only if it outperforms the control memory by a meaningful margin.
4. All projector results in the paper are paired across identical evaluation rows.

D. HumanEval Problem-Level Passes

On the 20-problem HumanEval subset, the base model solves HumanEval/16 and HumanEval/18, while BM25+PLE solves HumanEval/0 and HumanEval/10. The two pass sets are disjoint. This supports the claim that PLE

provides a different source of local code knowledge rather than simply amplifying the base model.

E. Case Study: TriviaQA Failure

The frozen 0.8B model obtains zero exact match on the 100-example TriviaQA subset. The failure is systematic rather than a calibration artifact: short-form knowledge questions require world knowledge that is not present in local n-gram continuations. A raw PLE fusion cannot recover this knowledge and may instead produce plausible but incorrect continuations.

F. Case Study: Open-Ended Degradation

Without the token-level policy, unconditional PLE fusion on natural-language code prompts produces repetitive fragments, unrelated repository text, and broken structure. The learned policy suppresses PLE on tokens where the n-gram prior is not clearly beneficial, which substantially reduces this degradation.

G. Limitations and Future Work

1. HumanEval is limited to 20 problems in the current version.
2. TriviaQA exact match is zero, so the paper reports a boundary rather than a positive result.
3. Public weights have not yet been released.
4. CPU latency and quantization remain unoptimized.
5. Future work includes memory-size scaling, n-gram-order ablations, and full joint-system evaluation on larger real benchmarks.

References

- Asai, A., Wu, Z., Wang, Y., Sil, A., and Hajishirzi, H. *Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*, 2023. <https://arxiv.org/abs/2310.11511v1>
- Asl, M. A., Asgari-Bidhendi, M., and Minaei-Bidgoli, B. *FAIR-RAG: Faithful Adaptive Iterative Refinement for Retrieval-Augmented Generation*, 2025. <https://arxiv.org/abs/2510.22344v1>
- Authors, R. *RAGRouter-Bench: Benchmarks for Learned Query Routing in Retrieval-Augmented Generation*, 2026. <https://arxiv.org/html/2602.00296>
- Authors, T. *TokenMem: Faithful Knowledge Injection for Frozen LLMs*, 2026. <https://arxiv.org/html/2607.22625>
- Bai, Y., Lv, X., Zhang, J., and others. *LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding*, 2024. <https://arxiv.org/abs/2308.14508>
- Berges, V.-P., Oguz, B., Haziza, D., and others. *Memory Layers at Scale. International Conference on*

- Machine Learning*, 2025. <https://mlanthology.org/icml/2025/berges2025icml-memory/>
- Bhardwaj, R., Vaddamanu, S., and others. kNN-LM Does Not Improve Open-ended Text Generation. *Proceedings of EMNLP*, 2023. <https://aclanthology.org/2023.emnlp-main.929/>
- Blackwell, D. *Comparison of Experiments*. University of California Press, 1951.
- Borro, L. C., Macarini, L. A. B., Tindall, G., Montero, M., and Struck, A. B. *Memori: A Persistent Memory Layer for Efficient, Context-Aware LLM Agents*, 2026. <https://arxiv.org/abs/2603.19935v1>
- Chen, M., Tworek, J., Jun, H., and others. *Evaluating Large Language Models Trained on Code*, 2021. <https://arxiv.org/abs/2107.03374>
- Cheng, R. et al. *Memory Grafting: Scaling Language Model Pre-training via Offline Conditional Memory*, 2026. <https://papers.cool/arxiv/2605.20948>
- Cheng, X. et al. *Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models*, 2026. <https://github.com/deepseek-ai/Engram>
- Cobbe, K., Kosaraju, V., Bavarian, M., and others. *Training Verifiers to Solve Math Word Problems*, 2021. <https://arxiv.org/abs/2110.14168>
- Das, P., Ko, C.-Y., Dai, S., Kollias, G., Chaudhury, S., and Lozano, A. *Can Memory-Augmented Language Models Generalize on Reasoning-in-a-Haystack Tasks?*, 2025. <https://arxiv.org/abs/2503.07903v1>
- Dettmers, T., Pagnoni, A., Holtzman, A., and Zettlemoyer, L. QLoRA: Efficient Finetuning of Quantized LLMs. *Advances in Neural Information Processing Systems*, 2023.
- Genest, C., and Zidek, J. V. Combining Probability Distributions: A Critique and an Annotated Bibliography. *Statistical Science*, 1986.
- Gros, D., and Devanbu, P. *Localized Calibrated Uncertainty in Code Language Models*, 2025. <https://arxiv.org/abs/2512.24560v1>
- Group, L. *MemSFT: Mitigating Alignment Tax with an External Parametric Memory*, 2026. <https://github.com/LUMIA-Group/MemSFT>
- Gutiérrez, B. J., Shu, Y., Gu, Y., Yasunaga, M., and Su, Y. *HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models*, 2024. <https://arxiv.org/abs/2405.14831v3>
- Gutiérrez, B. J., Shu, Y., Qi, W., Zhou, S., and Su, Y. *From RAG to Memory: Non-Parametric Continual Learning for Large Language Models*, 2025. <https://arxiv.org/abs/2502.14802v2>
- He, J., Neubig, G., and Berg-Kirkpatrick, T. *Efficient Nearest Neighbor Language Models*, 2021. <https://arxiv.org/abs/2109.04212v3>
- Hendrycks, D., Burns, C., Kadavath, S., and others. *Measuring Mathematical Problem Solving with the MATH Dataset*, 2021. <https://arxiv.org/abs/2103.03874>
- Hu, E. J., Shen, Y., Wallis, P., and others. LoRA: Low-Rank Adaptation of Large Language Models. *International Conference on Learning Representations*, 2022.
- Huang, Y., Liu, Y., Zhao, R., Zhong, X., Yue, X., and Jiang, L. *MemOrb: A Plug-and-Play Verbal-Reinforcement Memory Layer for E-Commerce Customer Service*, 2025. <https://arxiv.org/abs/2509.18713v1>
- Hwang, J., Park, J., Park, H., Kim, D., Park, S., and Ok, J. *Retrieval-Augmented Generation with Estimation of Source Reliability*, 2024. <https://arxiv.org/abs/2410.22954v5>
- Jiang, J., Wu, X., Song, W., Yang, B., Zhou, Y., Ge, H., Lee, H. P., Liang, Y., and Wu, C. *ReliableRAG: Combating Misinformation in Retrieval-Augmented Generation via Reliability-Guided Reasoning Chains*, 2026. <https://arxiv.org/abs/2608.25487v1>
- Jiang, T., Huang, S., Luo, S., and others. *MoRA: High-Rank Updating for Parameter-Efficient Fine-Tuning*, 2024. <https://arxiv.org/abs/2405.12130>
- Jin, M., Luo, W., Cheng, S., Wang, X., Hua, W., Tang, R., Wang, W. Y., and Zhang, Y. *Disentangling Memory and Reasoning Ability in Large Language Models*, 2024. <https://arxiv.org/abs/2411.13504v3>
- Joshi, M., Choi, E., Weld, D., and Zettlemoyer, L. *TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension*, 2017. <https://aclanthology.org/P17-1147/>
- Khandelwal, U., Levy, O., Jurafsky, D., Zettlemoyer, L., and Lewis, M. Generalization through Memorization: Nearest Neighbor Language Models. *International Conference on Learning Representations*, 2020. <https://papers.lunadong.com/paper/4555>
- Kim, H., Kim, Y. J., and Bak, J. *PEMA: An Offsite-Tunable Plug-in External Memory Adaptation for Language Models*, 2023. <https://arxiv.org/abs/2311.08590v3>

- Lample, G., Sablayrolles, A., Ranzato, M., Denoyer, L., and Jégou, H. *Large Memory Layers with Product Keys*, 2019. <https://arxiv.org/abs/1907.05242v2>
- Lewis, P., Perez, E., Piktus, A., and others. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 2020.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., and Liang, P. *Lost in the Middle: How Language Models Use Long Contexts*, 2023. <https://arxiv.org/abs/2307.03172v3>
- Lyu, Q., Shridhar, K., Malaviya, C., Zhang, L., Elazar, Y., Tandon, N., Apidianaki, M., Sachan, M., and Callison-Burch, C. *Calibrating Large Language Models with Sample Consistency*, 2024. <https://arxiv.org/abs/2402.13904v2>
- Miao, M. M., and Ungar, L. *Closing the Confidence-Faithfulness Gap in Large Language Models*, 2026. <https://arxiv.org/abs/2603.25052v2>
- Nelson, E., Kollias, G., Das, P., Chaudhury, S., and Dan, S. *Needle in the Haystack for Memory Based Large Language Models*, 2024. <https://arxiv.org/abs/2407.01437v2>
- OLAResearch. *Cross-Model Memory Transfer via Target-Side Reader Adaptation*, 2026. <https://github.com/OLAResearch/XMemTransfer>
- Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S. G., Stoica, I., and Gonzalez, J. E. *MemGPT: Towards LLMs as Operating Systems*, 2023. <https://arxiv.org/abs/2310.08560v2>
- Pan, X., Yao, W., Zhang, H., Yu, D., Yu, D., and Chen, J. *Knowledge-in-Context: Towards Knowledgeable Semi-Parametric Language Models*, 2022. <https://arxiv.org/abs/2210.16433v3>
- PioneerQyw. *NGM: A Plug-and-Play Training-Free Memory Module for LLMs*, 2026. <https://github.com/PioneerQyw/NGM>
- Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., and Chalef, D. *Zep: A Temporal Knowledge Graph Architecture for Agent Memory*, 2025. <https://arxiv.org/abs/2501.13956v1>
- Robertson, S., and Zaragoza, H. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 2009.
- Shin, S., Cha, M., Lee, B.-J., and Park, S. *ERA: Evidence-based Reliability Alignment for Honest Retrieval-Augmented Generation*, 2026. <https://arxiv.org/abs/2604.20854v1>
- Sutton, R. *The Bitter Lesson*, 2019. <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>
- Wang, H., Luo, Y., and others. *MemTrapBench: Benchmarking Cognitive Traps in LLM Memory Use*, 2026. <https://www.semanticscholar.org/paper/MemTrapBench%3A-Benchmarking-Cognitive-Traps-in-LLM-Wang-Luo/736d61825a5afed4c85b227951a9880d01e2299f>
- Xu, F. F., Alon, U., and Neubig, G. *Why do Nearest Neighbor Language Models Work?*, 2023. <https://arxiv.org/abs/2301.02828v2>
- Xu, J., Shi, H., Shan, Y., Liu, P., Bai, Y., Li, N., and Liu, X. *TrustMargin: Training-Free Arbitration between Parametric Memory and Retrieved Evidence in Large Language Models*, 2026. <https://arxiv.org/abs/2606.08397v1>
- Yu, Z., Xing, W., Wei, Y., Yang, B., Ye, C., Li, G., and Han, M. *The Attribution Blind Spot: Detecting When Language Models Rely on Memory Rather Than Retrieved Context*, 2026. <https://arxiv.org/abs/2605.26778v1>
- Yu, Z., Zhao, Y., Cohan, A., and Zhang, X.-P. *HumanEval Pro and MBPP Pro: Evaluating Large Language Models on Self-invoking Code Generation*, 2024. <https://arxiv.org/abs/2412.21199v2>
- Zhang, Z., Zeng, Z., Lin, Y., Wang, H., Ye, D., Xiao, C., Han, X., Liu, Z., Li, P., Sun, M., and Zhou, J. *Plug-and-Play Knowledge Injection for Pre-trained Language Models*, 2023. <https://arxiv.org/abs/2305.17691v2>
- Zheng, L., Chiang, W.-L., Sheng, Y., and others. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*, 2024. <https://arxiv.org/abs/2306.05685>
- Zheng, Y., Wen, B., and Wang, X. *Lngram v2: Latent N-Gram Memory with Interpretable Discrete Representations*, 2026a. <https://arxiv.org/abs/2609.03426v1>
- Zheng, Y., Xia, G., Wang, X., and Ren, L. *Lngram: N-gram Conditional Memory in Latent Space*, 2026b. <https://arxiv.org/abs/2605.24869v1>
- Zhong, Z., Lei, T., and Chen, D. *Training Language Models with Memory Augmentation*, 2022. <https://arxiv.org/abs/2205.12674v3>
- Zhou, W., Gu, Y., Iacovides, G., Qiu, Y., Zhao, Q., and Mandic, D. *Tensorizing Engram: Sharing Latents Across N-Gram Embeddings is Beneficial in LLMs*, 2026. <https://arxiv.org/abs/2606.08347v1>